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# **MCQ Questions & Answers**

## **LinearMind Platform**

82 Questions across 12 Modules

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## 1 Introduction (6 Questions)

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**Q1. What does Linear Regression try to find?**

- a) The average of all data points
- b) The best straight line that fits the data
- c) The maximum value in the dataset
- d) Clusters in the data

**Q2. Which type of problem is Linear Regression designed to solve?**

- a) Classification: assigning categories
- b) Regression: predicting continuous values
- c) Clustering: grouping similar items
- d) Ranking: ordering items

**Q3. You want to predict house prices based on square footage. Which approach fits best?**

- a) K-Means Clustering
- b) Linear Regression
- c) Decision Tree Classification
- d) Image recognition with CNN

**Q4. On a scatter plot, what does a good linear regression fit look like?**

- a) All points stacked vertically
- b) Points scattered randomly with no pattern
- c) Points roughly following a straight line with the regression line passing through them
- d) Points forming a perfect circle

**Q5. You have a dataset with 3 features: temperature, humidity, and “temperature  $\times$  2”. You apply multiple linear regression. What problem will you encounter?**

- a) The model will train faster

- b) Perfect multicollinearity: the design matrix is singular and the closed-form solution fails
- c) The model will overfit
- d) The MSE will be negative

**Q6. Linear Regression assumes a linear relationship between inputs and outputs. Which of these relationships can it still model correctly?**

- a)  $y = \sin(x)$
- b)  $y = 3x_1^2 + 2x_2$  (if  $x_1^2$  is given as a feature)
- c)  $y = e^x$
- d)  $y = x_1 \times x_2$  (interaction without feature engineering)

## 2 The Mathematics (7 Questions)

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**Q7. In the equation  $y = mx + b$ , what does “m” represent?**

- a) The y-intercept
- b) The mean of the data
- c) The slope of the line
- d) The number of data points

**Q8. If  $y = 3x + 2$ , what is the predicted value when  $x = 4$ ?**

- a) 10
- b) 12
- c) 14
- d) 16

**Q9. If the slope of a regression line is negative, what does that tell us?**

- a) There is no relationship between x and y
- b) As x increases, y also increases
- c) As x increases, y decreases

- d) The model has failed

**Q10. If  $y = -2x + 10$ , what is  $y$  when  $x = 3$ ?**

- a) 4
- b) 16
- c) -4
- d) 6

**Q11. What does the y-intercept (b) represent on a graph?**

- a) The steepness of the line
- b) Where the line crosses the x-axis
- c) Where the line crosses the y-axis (when  $x = 0$ )
- d) The average of all  $y$  values

**Q12. The normal equation for linear regression is  $\mathbf{w} = (X^T X)^{-1} X^T \mathbf{y}$ . If  $X$  is a  $100 \times 3$  matrix, what is the shape of  $\mathbf{w}$ ?**

- a)  $100 \times 1$
- b)  $3 \times 1$
- c)  $3 \times 100$
- d)  $1 \times 3$

**Q13. Why does the closed-form solution (normal equation) become impractical for very large datasets?**

- a) It requires more training epochs
- b) Inverting  $X^T X$  has  $O(n^3)$  complexity in the number of features, which is too slow for high-dimensional data
- c) It always underfits
- d) It cannot handle negative values

### 3 Cost Function (8 Questions)

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**Q14. Why do we SQUARE the errors in MSE instead of just summing them?**

- a) To make computation faster
- b) Because squared numbers are always positive
- c) So positive and negative errors do not cancel out, and large errors are penalized more
- d) It is just a convention with no mathematical reason

**Q15. If you drag the regression line far away from the data points, what happens to the MSE?**

- a) It stays the same
- b) It decreases
- c) It increases dramatically
- d) It becomes negative

**Q16. An MSE of 0 means:**

- a) The model is terrible
- b) The model perfectly predicts every data point
- c) The model has not been trained yet
- d) There is a bug in the code

**Q17. Two models have MSE of 25 and 100 respectively. Which is better and by how much?**

- a) The second model is better because  $100 > 25$
- b) The first model is 4x better: lower MSE means less error
- c) They are equally good
- d) Cannot be determined from MSE alone

**Q18. What is the “residual” in regression?**

- a) The slope of the line

- b) The difference between the actual value and the predicted value
- c) The learning rate
- d) The total number of data points

**Q19. MSE uses squared errors. MAE uses absolute errors. When would you prefer MAE over MSE?**

- a) When you want to penalize large errors more
- b) When your data has many outliers and you want the model to be robust to them
- c) When you need a differentiable loss function
- d) Always: MAE is strictly better

**Q20. Given points (1, 2), (2, 4), (3, 5) and the line  $y = 1.5x + 0.5$ , what is the MSE?**

- a) 0
- b) 0.0833
- c) 0.25
- d) 0.5

**Q21. You compute the gradient of MSE and find  $\frac{\partial L}{\partial w} = 0$  and  $\frac{\partial L}{\partial b} = 0$ . What does this tell you?**

- a) The model has not started training yet
- b) The current  $w$  and  $b$  are at a critical point (minimum) of the loss function
- c) The learning rate is zero
- d) The data has no pattern

## 4 Gradient Descent (8 Questions)

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**Q22. What happens if the learning rate is TOO HIGH?**

- a) The model converges faster
- b) The model overshoots the minimum and may never converge
- c) The model underfits

- d) Nothing changes

**Q23. You are training a model and the loss keeps bouncing up and down without decreasing. What is the most likely cause?**

- a) The dataset is too small
- b) The learning rate is too high
- c) The model needs more features
- d) The bias term is wrong

**Q24. In Gradient Descent, the gradient tells us:**

- a) The exact location of the minimum
- b) The direction of steepest INCREASE of the loss
- c) The learning rate to use
- d) How many epochs are needed

**Q25. Your model is training very slowly, taking thousands of epochs. The loss decreases but extremely gradually. What should you do?**

- a) Decrease the learning rate
- b) Increase the learning rate
- c) Add more data points
- d) Remove the bias term

**Q26. On the cost function surface, what does the lowest point represent?**

- a) The worst possible model
- b) The starting point of training
- c) The optimal weights that minimize error
- d) The learning rate value

**Q27. What is the partial derivative of  $\text{MSE} = \frac{1}{n} \sum (wx_i + b - y_i)^2$  with respect to  $w$ ?**

- a)  $\frac{2}{n} \sum (wx_i + b - y_i)$



b)  $\frac{2}{n} \sum (wx_i + b - y_i) \cdot x_i$

c)  $\frac{1}{n} \sum x_i$

d)  $\frac{2}{n} \sum y_i$

**Q28. You plot the loss curve and notice it decreases, then increases, then decreases again in a zigzag pattern. What is the best fix?**

- a) Use a larger learning rate
- b) Use learning rate decay: start high and gradually reduce it
- c) Remove all features
- d) Use MSE instead of MAE

**Q29. For a single data point ( $x = 2, y = 6$ ) with current  $w = 1$  and  $b = 0$ , what is  $\partial \text{MSE} / \partial w$ ?**

- a) -16
- b) -8
- c) 8
- d) 16

## 5 Training Process (6 Questions)

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**Q30. What is one “epoch” in the training process?**

- a) One update to the weights
- b) One complete pass through the entire dataset
- c) One prediction
- d) One gradient calculation

**Q31. After 10 epochs the loss is 500. After 100 epochs it is 50. After 200 epochs it is 49.8. What should you do?**

- a) Keep training for 1000 more epochs
- b) Stop training: the model has converged

- c) Increase the learning rate dramatically
- d) Start over with random weights

**Q32. What happens to the weights during each training step?**

- a) They are randomly reassigned
- b) They are updated in the direction that reduces the loss
- c) They stay the same until the final epoch
- d) They are always increased

**Q33. If weight = 2.0, gradient = 0.5, learning rate = 0.1, what is the new weight?**

- a) 2.05
- b) 1.95
- c) 2.5
- d) 1.5

**Q34. What is the difference between batch, mini-batch, and stochastic gradient descent?**

- a) They use different loss functions
- b) Batch uses all data per update, mini-batch uses a subset, SGD uses a single sample
- c) SGD is always faster and more accurate
- d) There is no practical difference

**Q35. A model trained for 50 epochs has training MSE=10 and test MSE=100. What is happening?**

- a) The model is underfitting
- b) The model is overfitting: it memorized training data but does not generalize
- c) The model is perfectly trained
- d) The test set is too large

## 6 Failure Cases (7 Questions)

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**Q36. A dataset contains a clear quadratic (curved) relationship. You apply Linear Regression. What happens?**

- a) It fits perfectly
- b) It underfits: the straight line cannot capture the curve
- c) It overfits
- d) The MSE becomes zero

**Q37. A dataset has one extreme outlier. What effect does it have on the regression line?**

- a) No effect at all
- b) It pulls the line toward the outlier, distorting predictions for other points
- c) It improves the model accuracy
- d) It only affects the intercept

**Q38. Adding heavy random noise to a dataset will likely cause:**

- a) The MSE to decrease
- b) Perfect predictions
- c) Higher MSE and less reliable predictions
- d) The model to overfit

**Q39. What is “multicollinearity” and why is it a problem?**

- a) When features are unrelated: makes the model too simple
- b) When two or more features are highly correlated: makes weight estimates unstable
- c) When there is only one feature: limits accuracy
- d) When the target variable has multiple values

**Q40. If your data forms a U-shape (parabola), what will the regression line look like?**

- a) A perfect curve matching the data

- b) A horizontal line cutting through the middle, missing both ends
- c) A vertical line
- d) Two separate lines

**Q41. Your linear regression has  $R^2 = 0.15$ . What does this mean?**

- a) The model explains 85% of variance: it is great
- b) The model explains only 15% of variance: the relationship is likely nonlinear or key features are missing
- c)  $R^2$  of 0.15 is always acceptable
- d) The model has converged optimally

**Q42. What is heteroscedasticity?**

- a) When the mean of errors is non-zero
- b) When the variance of residuals changes across different values of  $x$ , violating the constant-variance assumption
- c) When errors are correlated with each other
- d) When the data has too few points

## 7 Comparison (7 Questions)

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**Q43. What is the key advantage of Linear Regression over Neural Networks?**

- a) It is more accurate
- b) It can handle nonlinear data
- c) It is simpler, faster, and more interpretable
- d) It requires more data

**Q44. Which method should you try FIRST for a dataset with a clear linear trend?**

- a) Deep Neural Network
- b) Random Forest
- c) Linear Regression

d) Convolutional Neural Network

**Q45. You have 50 data points with a clear linear trend. A colleague suggests a neural network with 10 layers. What do you recommend?**

- a) Agree: deeper is always better
- b) Use Linear Regression: it is sufficient and will not overfit
- c) Use at least 20 layers
- d) Do not use any model

**Q46. What does “interpretability” mean in the context of models?**

- a) How fast the model trains
- b) How easily humans can understand WHY the model makes its predictions
- c) How many parameters the model has
- d) The programming language used

**Q47. Your image recognition task needs to classify cats vs dogs. Should you use Linear Regression?**

- a) Yes: it works for everything
- b) No: this is a classification task with complex, nonlinear patterns requiring a different approach
- c) Yes, but only with a high learning rate
- d) Only if you have less than 100 images

**Q48. What is regularization (L1/L2)?**

- a) It increases the learning rate for faster training
- b) It adds a penalty term to the loss to prevent weights from becoming too large, reducing overfitting
- c) It removes outliers from the dataset
- d) It converts regression into classification

**Q49. You apply Ridge Regression (L2) with  $\lambda = 1000$ . What happens to the weights?**

- a) They become very large
- b) They are pushed very close to zero, severely underfitting the data
- c) They are unaffected
- d) They become negative

## 8 Multivariate Regression (13 Questions)

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**Q50. What does multivariable linear regression extend compared to simple linear regression?**

- a) The number of output variables
- b) The number of input features
- c) The number of training epochs
- d) The number of loss functions

**Q51. In the equation  $\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 + b$ , how many features does the model use?**

- a) 1
- b) 2
- c) 3
- d) 4

**Q52. You are predicting house prices using size, bedrooms, and age. Which is the correct model form?**

- a)  $\hat{y} = w \cdot \text{size} + b$
- b)  $\hat{y} = w_1 \cdot \text{size} + w_2 \cdot \text{bedrooms} + w_3 \cdot \text{age} + b$
- c)  $\hat{y} = w \cdot (\text{size} + \text{bedrooms} + \text{age}) + b$
- d)  $\hat{y} = w_1 \cdot \text{size} \times w_2 \cdot \text{bedrooms} \times w_3 \cdot \text{age} + b$

**Q53. In the matrix formulation  $\hat{Y} = XW + b$ , what are the dimensions of  $X$  if there are 100 samples and 5 features?**

- a)  $5 \times 100$

- b)  $100 \times 5$
- c)  $100 \times 100$
- d)  $5 \times 5$

**Q54. What is the Normal Equation for finding optimal weights?**

- a)  $W^* = X^T y$
- b)  $W^* = (X^T X)^{-1} X^T y$
- c)  $W^* = X^{-1} y$
- d)  $W^* = (X X^T)^{-1} X y$

**Q55. Feature “size” ranges from 50–500 while “bedrooms” ranges from 1–6. What problem does this cause?**

- a) The model cannot converge
- b) Gradient descent zig-zags and converges slowly
- c) The bias becomes negative
- d) The model always overfits

**Q56. What does Min-Max Normalization scale features to?**

- a)  $[-1, 1]$
- b)  $[0, 1]$
- c) Mean= 0, Std= 1
- d)  $[0, 100]$

**Q57. What does Z-score standardization produce?**

- a) Values between 0 and 1
- b) Values between  $-1$  and 1
- c) Mean = 0 and standard deviation = 1
- d) All positive values

**Q58. What is feature engineering?**

- a) Removing all features from the dataset
- b) Creating new features from existing ones to improve predictions
- c) Training a model without any features
- d) Using only one feature at a time

**Q59. You have 50,000 features. Should you use the Normal Equation or Gradient Descent?**

- a) Normal Equation: it is always exact
- b) Gradient Descent: Normal Equation is  $O(n^3)$  and too slow for many features
- c) Either works equally well
- d) Neither: use a neural network instead

**Q60. What is multicollinearity and why is it problematic?**

- a) When features are uncorrelated with the target
- b) When features are highly correlated with each other, causing unstable weights
- c) When the model has too many layers
- d) When the learning rate is too high

**Q61. Given  $\hat{y} = 3x_1 + 2x_2 - x_3 + 5$ , what is the prediction for  $x_1 = 2, x_2 = 4, x_3 = 1$ ?**

- a) 16
- b) 18
- c) 14
- d) 20

**Q62. Your features have very different scales:  $x_1 \in [0, 1]$  and  $x_2 \in [0, 1000000]$ . You run gradient descent without normalization. What happens?**

- a) Nothing: gradient descent handles all scales equally
- b) The loss surface becomes elongated, causing slow zigzag convergence
- c) The model only learns  $x_2$  and ignores  $x_1$
- d) The MSE becomes negative



## 9 Data Intuition (5 Questions)

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**Q63. What does a Pearson correlation coefficient of  $r = 0$  mean?**

- a) Perfect positive relationship
- b) No linear relationship
- c) Perfect negative relationship
- d) The data has errors

**Q64. If data points form a tight cluster along a downward-sloping line, what is the likely correlation?**

- a)  $r \approx +0.9$
- b)  $r \approx -0.9$
- c)  $r \approx 0$
- d)  $r \approx +0.5$

**Q65. Ice cream sales and drowning deaths have  $r \approx 0.85$ . Does this mean ice cream causes drowning?**

- a) Yes, the high correlation proves causation
- b) No: both are caused by a confounding variable (hot weather)
- c) Only if  $r > 0.95$
- d) This correlation is impossible

**Q66. Which scenario would produce a Pearson  $r \approx 0$  despite a clear pattern in the data?**

- a) A perfectly linear positive trend
- b) A U-shaped (quadratic) relationship
- c) A moderate negative trend
- d) A strong positive trend with noise

**Q67. If  $r = -1.0$ , what does the scatter plot look like?**

- a) Random scatter

- b) All points exactly on an upward line
- c) All points exactly on a downward line
- d) A circular pattern

## 10 Bias-Variance Tradeoff (5 Questions)

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**Q68. What is “underfitting” in machine learning?**

- a) The model memorizes training data
- b) The model is too simple to capture the underlying pattern
- c) The model works perfectly on test data
- d) The model has too many parameters

**Q69. What happens when a model overfits?**

- a) It performs well on both train and test data
- b) It has high training error
- c) It memorizes training data but fails on new data
- d) It is too simple

**Q70. Your model has train MSE = 0.5 and test MSE = 25. What is the diagnosis?**

- a) Underfitting
- b) Good fit
- c) Overfitting
- d) Insufficient data

**Q71. In the bias-variance tradeoff, increasing model complexity typically:**

- a) Increases bias and decreases variance
- b) Decreases bias and increases variance
- c) Decreases both bias and variance
- d) Increases both bias and variance

**Q72. A degree-15 polynomial is fit to 20 data points. What will likely happen?**

- a) Perfect generalization
- b) High bias, low variance
- c) Low bias, high variance (overfitting)
- d) The model cannot be trained

## 11 Evaluation Metrics (5 Questions)

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**Q73. What does  $R^2 = 1.0$  mean?**

- a) The model has 100% error
- b) The model explains all variance in the target
- c) The model is overfitting
- d) The features are uncorrelated

**Q74. Which metric is MOST sensitive to outliers?**

- a) MAE (Mean Absolute Error)
- b) MSE (Mean Squared Error)
- c) Median Error
- d)  $R^2$

**Q75. Predictions: [10, 12, 15]. Actual: [11, 10, 14]. What is the MAE?**

- a) 1.0
- b) 1.33
- c) 2.0
- d) 1.67

**Q76. Your model has  $RMSE = 5.0$  predicting house prices in thousands. What does this mean practically?**

- a) Predictions are off by \$5 on average

- b) Predictions are off by about \$5,000 on average
- c) The model explains 5% of variance
- d) The model has 5 features

**Q77. Can  $R^2$  be negative? If so, when?**

- a) No,  $R^2$  is always between 0 and 1
- b) Yes: when the model is worse than predicting the mean
- c) Yes: only for nonlinear models
- d) Yes: when there are too many features

## 12 Linear Regression to Neural Networks (5 Questions)

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**Q78. A single neuron with no activation function is equivalent to:**

- a) A decision tree
- b) Linear regression
- c) A random forest
- d) Logistic regression

**Q79. Why do neural networks need activation functions?**

- a) To make training faster
- b) To introduce non-linearity so the network can model complex patterns
- c) To reduce the number of parameters
- d) To prevent overfitting

**Q80. What does the ReLU activation function do?**

- a) Outputs the input if positive, 0 otherwise
- b) Squishes values between 0 and 1
- c) Squares the input
- d) Normalizes the output

**Q81. If you stack 3 linear layers without activations ( $y = W_3(W_2(W_1x + b_1) + b_2) + b_3$ ), what do you get?**

- a) A deep neural network
- b) A single linear transformation (still linear regression)
- c) A cubic function
- d) A model that cannot be trained

**Q82. A neuron has weights  $[2, -1]$ , bias = 3, and ReLU activation. Input =  $[1, 5]$ . What is the output?**

- a) 0
- b) 6
- c) -2
- d) 0 (because ReLU clips negative values)

## 13 Answer Key

All correct answers with brief explanations.

| Q# | Ans | Explanation                                                                                                                        |
|----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| 1  | b   | Linear Regression finds the best straight line ( $y = wx + b$ ) that minimizes the distance between predictions and actual values. |
| 2  | b   | It predicts continuous numeric values (e.g., price, temperature) rather than discrete categories.                                  |
| 3  | b   | Predicting a continuous value (price) from a numeric feature (square footage) is a classic Linear Regression problem.              |
| 4  | c   | A good fit shows points roughly following a linear trend, with the regression line passing through the center.                     |
| 5  | b   | The third feature is an exact linear combination of the first, making $(X^T X)$ singular.                                          |
| 6  | b   | Linear regression is linear in its parameters, not its features. If $x_1^2$ is provided as a feature, the model fits it perfectly. |
| 7  | c   | The slope (m) determines how steep the line is.                                                                                    |
| 8  | c   | $y = 3(4) + 2 = 14$                                                                                                                |
| 9  | c   | A negative slope means as x increases, y decreases.                                                                                |
| 10 | a   | $y = -2(3) + 10 = 4$                                                                                                               |
| 11 | c   | The y-intercept is the value of y when $x = 0$ .                                                                                   |
| 12 | b   | $X^T X$ is $3 \times 3$ , so $\mathbf{w} = (3 \times 3)(3 \times 1) = 3 \times 1$ .                                                |
| 13 | b   | Inverting $X^T X$ has $O(p^3)$ complexity where p is the number of features.                                                       |
| 14 | c   | Squaring prevents cancellation of positive/negative errors and penalizes larger errors more.                                       |
| 15 | c   | Moving the line away increases squared errors dramatically.                                                                        |
| 16 | b   | $\text{MSE} = 0$ means every prediction exactly matches the actual value.                                                          |
| 17 | b   | Lower MSE means less error. The first model has 4x less error.                                                                     |

| Q# | Ans | Explanation                                                                                   |
|----|-----|-----------------------------------------------------------------------------------------------|
| 18 | b   | Residual = actual – predicted.                                                                |
| 19 | b   | MAE is more robust to outliers since it does not square errors.                               |
| 20 | b   | Predictions: 2, 3.5, 5. Errors <sup>2</sup> : 0, 0.25, 0. MSE = $0.25/3 = 0.0833$ .           |
| 21 | b   | Zero gradients mean you are at the global minimum (MSE is convex for linear regression).      |
| 22 | b   | Too-high learning rate causes overshooting and divergence.                                    |
| 23 | b   | Oscillating loss typically indicates a learning rate that is too large.                       |
| 24 | b   | The gradient points in the direction of steepest increase; we move opposite to decrease loss. |
| 25 | b   | Very slow convergence means the learning rate is too small.                                   |
| 26 | c   | The lowest point is the optimal weights that minimize error.                                  |
| 27 | b   | By chain rule: $\frac{2}{n} \sum (wx_i + b - y_i) \cdot x_i$ .                                |
| 28 | b   | Learning rate decay starts high for fast progress, then reduces for stable convergence.       |
| 29 | a   | $\hat{y} = 2$ , error = -4. $\partial \text{MSE} / \partial w = 2(-4)(2) = -16$ .             |
| 30 | b   | An epoch is one complete pass through all training data.                                      |
| 31 | b   | Loss barely changed ( $50 \rightarrow 49.8$ ): the model has converged.                       |
| 32 | b   | Weights are updated in the direction that reduces the loss.                                   |
| 33 | b   | $2.0 - 0.1 \times 0.5 = 1.95$                                                                 |
| 34 | b   | Batch uses all data, mini-batch a subset, SGD one sample per update.                          |
| 35 | b   | Large gap between train and test error is overfitting.                                        |
| 36 | b   | A straight line cannot capture a curve: underfitting.                                         |
| 37 | b   | MSE penalizes large errors, pulling the line toward the outlier.                              |
| 38 | c   | Noise obscures the true pattern, raising MSE.                                                 |

| Q# | Ans | Explanation                                                                        |
|----|-----|------------------------------------------------------------------------------------|
| 39 | b   | Highly correlated features make weight estimates unstable.                         |
| 40 | b   | A straight line cuts through the middle of a U-shape, missing both ends.           |
| 41 | b   | Only 15% of variance is explained; likely nonlinear or missing features.           |
| 42 | b   | Error variance changes with $x$ , violating the constant-variance assumption.      |
| 43 | c   | Linear Regression is simpler, faster, and more interpretable.                      |
| 44 | c   | Always start simple with Linear Regression for linear data.                        |
| 45 | b   | For 50 points with a linear trend, LR is sufficient; a deep network would overfit. |
| 46 | b   | How easily humans can understand why the model makes its predictions.              |
| 47 | b   | Image classification needs a CNN, not regression.                                  |
| 48 | b   | Regularization penalizes large weights to reduce overfitting.                      |
| 49 | b   | Very high $\lambda$ pushes all weights toward zero, causing underfitting.          |
| 50 | b   | Extends from one input feature to multiple input features.                         |
| 51 | c   | Three features: $x_1, x_2, x_3$ . $b$ is the bias.                                 |
| 52 | b   | Each feature gets its own independent weight.                                      |
| 53 | b   | $X$ is (samples $\times$ features) = $100 \times 5$ .                              |
| 54 | b   | $W^* = (X^T X)^{-1} X^T y$                                                         |
| 55 | b   | Different scales cause elongated contours and slow zig-zag convergence.            |
| 56 | b   | Maps values to $[0, 1]$ using $(x - \min)/(\max - \min)$ .                         |
| 57 | c   | Centers at mean = 0 with standard deviation = 1.                                   |
| 58 | b   | Creating new features from existing ones (e.g., $x^2$ , price/size).               |
| 59 | b   | Normal Equation is $O(n^3)$ ; gradient descent scales better.                      |
| 60 | b   | Correlated features cause unstable weight estimates.                               |



| Q# | Ans | Explanation                                                                 |
|----|-----|-----------------------------------------------------------------------------|
| 61 | b   | $3(2) + 2(4) - 1(1) + 5 = 18$                                               |
| 62 | b   | Unscaled features create an elongated loss surface.                         |
| 63 | b   | No linear relationship between the variables.                               |
| 64 | b   | Tight downward cluster = strong negative correlation.                       |
| 65 | b   | Correlation $\neq$ causation; hot weather is the confounding variable.      |
| 66 | b   | U-shaped patterns cancel out, giving $r \approx 0$ despite a clear pattern. |
| 67 | c   | $r = -1.0$ : all points on a downward line.                                 |
| 68 | b   | The model is too simple to capture the pattern.                             |
| 69 | c   | Memorizes training data but fails on new data.                              |
| 70 | c   | Huge train/test gap = overfitting.                                          |
| 71 | b   | More complexity = less bias, more variance.                                 |
| 72 | c   | 16 parameters for 20 points = extreme overfitting.                          |
| 73 | b   | The model explains all variance in the target.                              |
| 74 | b   | MSE squares errors, so outliers dominate.                                   |
| 75 | b   | $(1 + 2 + 1)/3 \approx 1.33$                                                |
| 76 | b   | RMSE is in target units: \$5,000 off on average.                            |
| 77 | b   | $R^2 < 0$ when the model is worse than predicting the mean.                 |
| 78 | b   | A neuron with no activation = linear regression.                            |
| 79 | b   | Activations introduce non-linearity for complex patterns.                   |
| 80 | a   | $\text{ReLU}(x) = \max(0, x)$ .                                             |
| 81 | b   | Stacked linear layers = one linear transformation.                          |
| 82 | a   | $z = 2(1) + (-1)(5) + 3 = 0$ . $\text{ReLU}(0) = 0$ .                       |